MATH2099 Notes

Contents

Python

Preamble

```python
import numpy as np
import scipy as sci
import pandas as pd
import matplotlib.pyplot as plt
```

Linear Algebra

Definitions

Vector definition	$\begin{pmatrix} x \\ y \\ z \\ \dots \end{pmatrix}$	`np.array([x, y, z, ...])`

Matrix definition	$\begin{pmatrix} v_1 & v_2 & \dots & v_n \\ v_4 & v_5 & \dots & v_6 \\ \vdots & \vdots & \ddots & \vdots \\ v_7 & v_8 & \dots & v_9 \end{pmatrix}$	`np.array([[v1, v2,  ..., v3],`
		`          [v4, v5,  ..., v6],`
		`          [..., ...,..., ...],`
		`          [v7, v8,  ..., v9]])`

Matrix augmentation	$(v_1 \mid v_2 \mid \dots \mid v_n)$	`np.column_stack((v1, v2, ..., v3))`
	$\begin{pmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{pmatrix}$	`np.row_stack((v1, v2, ..., v3))`
Zero matrix / vector	$\begin{pmatrix} 0 \\ 0 \\ \dots \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}$	`np.zeros(n)` `np.zeros((m, n))`
Identity matrix	$\boldsymbol{I} \in \mathbb{C}^n$	`np.eye(n)`
Diagonal matrix	$\begin{pmatrix} a & 0 & \dots & 0 \\ 0 & b & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & c \end{pmatrix}$	`np.diag([a, b, ..., c])`

Vector element access	v_i	`v[i] # 0-indexed`
Matrix row access	$\boldsymbol{M}_i$	`M[i] # 0-indexed`
Matrix column access	$\boldsymbol{M}_{:,i}$	`M[:, i] # 0-indexed`
Matrix element access	$\boldsymbol{M}_{ij}$	`M[i, j] # 0-indexed`

Vector dot product	$a \cdot b$	`np.dot(a, b)` or `a.dot(b)`
Matrix multiplication	$\boldsymbol{AB}$	`A @ B`
Matrix determinate	$\det(\boldsymbol{A})$	`np.linalg.det(A)`
Matrix inverse	$\mathrm{inv}(\boldsymbol{A})$	`np.linalg.inv(A)`
Matrix transpose	$\boldsymbol{A}^T$	`A.T`
Solve linear equation	$\boldsymbol{A}x = b$	`x = np.solve(A, b)`
Eigen value / vector	$\mathrm{eig}(\boldsymbol{A})$	`np.linalg.eig(A)` `np.linalg.eigvals(A)`

Matrix rank	$\text{rank}(\boldsymbol{A})$	`np.linalg.matrix_rank(A)`
Matrix nullspace	$\text{nullspace}(\boldsymbol{A})$	`sci.linalg.null_space(A)`
Matrix norm	$\text{norm}(\boldsymbol{A})$	`np.linalg.norm(A, ord=?)`

Statistics

Import Data	`pd.read_csv("filename.txt", delimeter="\t")`
Mean	`np.mean(a)` `dataframe.mean()`
Median	`np.median(a)` `dataframe.median()`
Standard deviation	`np.std(a, ddof=1)` `dataframe.std()`
Variance	`np.var(a, ddof=1)` `dataframe.var()`

Quartiles	`np.quartiles(a,`
	`    [0, 0.25, 0.5, 0.75, 1], method='hazen')`
	`# For min, q1, median, q3, max`

Skewness	`sc.stats.skew(a)`

Plotting

Scatter Plot	`dataframe.plot.scatter(xcol, ycol)`
Line Plot	`dataframe.plot.line()`
Histogram	`dataframe.plot.hist(bins) # bins: int | []int`
Density Histogram	`dataframe.plot.hist(density=True, bins)`

Binomial $X \sim \text{Bin}(n, p)$ $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$ $E[X] = np$, $\text{Var}(X) = np(1-p)$	`sc.stats.binom(n, p)`
Poisson $X \sim \text{Poi}(\lambda)$ $P(X = k) = \lambda^k \frac{e^{-\lambda}}{k!}$ $E[X] = \lambda$, $\text{Var}(X) = \lambda$	`sc.stats.poisson(lam)`
Exponential $X \sim \text{Exp}(\lambda)$ $f(x) = \lambda e^{-\lambda x}$ $E[X] = \frac{1}{\lambda}$, $\text{Var}(X) = \frac{1}{\lambda^2}$	`sc.stats.expon(scale=1/lam)`
Normal $X \sim \mathcal{N}(\mu, \sigma^2)$ $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$ $E[X] = \mu$, $\text{Var}(X) = \sigma^2$	`sc.stats.norm(mu, sigma)`

Uniform
$X \sim \text{Uniform}(a, b)$
$f(x) = \frac{1}{b-a}$ for $x \in [a, b]$
$E[X] = \frac{a+b}{2}$, $\text{Var}(X) = \frac{(b-a)^2}{12}$

`sc.stats.uniform(loc=a, scale=b-a)`

Given `dist = sc.stats.xxx(...)`

`.pmf(k)` — $P(X = k)$ (discrete)
`.pdf(x)` — $f(x)$ (continuous)
`.cdf(x)` — $P(X \leq x)$
`.ppf(q)` — $F^{-1}(q)$ (quantile)
`.mean()`, `.var()`, `.std()`

Linear Algebra

Linear Equations and Matrices

Vector dot product

$$\vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i$$

Matrix Vector Multiplication

$$A\vec{v} = (a_1 \mid a_2 \mid \ldots \mid a_n)\vec{v} = a_1 v_1 + a_2 v_2 + \ldots + a_n v_n$$

Matrix Matrix Multiplication

$$AB = \left(A\vec{b}_1 \mid A\vec{b}_2 \mid \ldots \mid A\vec{b}_n\right)$$

Matrices Properties

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $A + 0 = A$
- $A + (-A) = 0$
- $\lambda(\mu A) = (\lambda\mu)A$
- $(\lambda + \mu)A = \lambda A + \mu A$
- $\lambda(A + B) = \lambda A + \lambda B$
- $A(BC) = (AB)C$
- $A(B + C) = AB + AC$
- $A(\lambda B) = \lambda(AB) = A(\lambda B)$
- $A0 = 0A = 0$
- $AI + IA + A$
- $AB \neq BA$(usually)

Transpose A^T = diagonally flipped from top left to bottom right
Inverse Matrix X is the inverse of A if $AX = XA = I$

Transpose and Inverse Properties:
- $\left(A^T\right)^T = A$
- $(\lambda A + \mu B)^T = \lambda A^T + \mu B^T$
- if AB exists, then $(AB)^T = B^T A^T$
- inverses only exist for square matrices $M \in C_{n,m}$ for $n = m$

- If A is invertible the $\left(A^{-1}\right)^{-1} = A$
- If both A and B are invertible and are the same shape, $(AB)^{-1} = B^{-1}A^{-1}$

Symmetric $A^T = A$
Skew-symmetric $A^T = -A$
Orthogonal $A^T A = AA^T \Leftrightarrow A^T = A^{-1}$

Vector Spaces

> **Vector Space** A **vector space** V **over the field of scalars** $\mathbb{F}$ is a non-empty set of **vectors** for which addition of vectors and multiplication of a vector by a scalar are defined and obey the following axioms.

1. Closure under addition. If $u, v \in V$ then $u + v \in V$.
2. Associative law of addition. If $u, v, w \in V$ then $(u + v) + w = u + (v + w)$.
3. Commutative law of addition. If $u, v \in V$ then $u + v = v + u$.
4. Existence of zero. There is a special element 0 in V called the **zero vector** which has the property that $v + 0 = v$ for all $v \in V$.
5. Existence of Netagive: For each $v \in V$ there exists an element $w \in V$ such that $v + w = 0$.
6. Closure under scalar multiplication. If $v \in V$ and $\lambda \in \mathbb{F}$, then $\lambda v \in V$.
7. Associative law of multiplication by a scalar. If $\lambda, \mu \in \mathbb{F}$ and $v \in V$ then $\lambda(\mu v) = (\lambda\mu)v$
8. Behaviour of scalar 1. If $v \in V$ then $1v = v$.
9. Scalar distributive law. If $\lambda, \mu \in \mathbb{F}$ and $v \in V$ then $(\lambda\mu)v = \lambda v + \mu v$.
10. Vector distributive law. If $\lambda \in \mathbb{F}$ and $u, v \in V$ then $\lambda(u + v) = \lambda u \wedge \lambda v$.

Important Vector Spaces

- The vector spaces $\mathbb{R}^n$ over $\mathbb{R}$, and $\mathbb{C}^n$ over $\mathbb{C}$.
- THe vector space of all polynomials of degree n or less with coefficients in $\mathbb{F}$, $\mathbb{P}_{n(\mathbb{F})}$ over $\mathbb{F}$.
- The set of all polynomials with coefficients in $\mathbb{F}$, $\mathbb{P}(\mathbb{F})$ over $\mathbb{F}$.
- The vector space of all $p \times q$ matrices, $M_{p,q}(\mathbb{F})$ over $\mathbb{F}$.
- The set of all real-values functions with domain X, $R[X]$ over $\mathbb{R}$.

Subspaces

> **Subspace** A subset S of a vector space V is called a **subset** of V is S is itself a vector space over the same field of scalars as V and under the same rules for addition and multiplication by scalars.

In addition, if S is a proper subset of V (i.e. $S \neq V$), S is called a **proper subspace** of V.

The set $\{0\}$ consisting of just the zero vector in V is always a subspace of V and is called the **trivial subspace** of V.

Proving Subspace (Subspace Theorem)

A subset S of a vector space V is a subspace if and only if:
- the zero vector of V belongs to S
- $u + v \in S$ for all $u, v \in S$
- $\lambda v \in S$ for all $v \in S$, scalars λ

Linear Combinations and Spans

> **Linear Combination Definition** Let $S = \{v_1, ..., v_n\}$ be a subset of a vector space V over a field $\mathbb{F}$. A **linear combination** of S is a sum of scalar multiples of the form
>
> $$\lambda_1 v_1 + ... + \lambda_n v_n \text{ with } \lambda_1, ..., \lambda_n \in \mathbb{F}$$

> **Span** Let $S = \{v_1, ..., v_n\}$ be a subset of a vector space V. Then the **span** of the set S is the set of all linear combinations of S, and is denoted by $\operatorname{span}(S)$ or $\operatorname{span}(v_1, ..., v_n)$.
>
> If $\operatorname{span}(S) = V$, the set S is called the **spanning set** of V, and S is said to **span** V or **generate** V.

Linear Independence

> **Linear Independence** Given a set of vectors $S = \{v_1, ..., v_n\}$. If the **only** solution of
>
> $$\lambda_1 v_1 + \lambda_2 v_2 + ... + \lambda_n v_n = 0$$
>
> is the **trivial one** i.e. $\lambda_1 = \lambda_2 = ... = \lambda_n = 0$, then we say that S is a **linearly independent set** and that the vectors in S are **linearly independent**.
>
> If we can find scalars $\lambda_1, \lambda_2, ..., \lambda_n$ **not all zero** such that
>
> $$\lambda_1 v_1 = \lambda_2 v_2 + ... + \lambda_n v_n = 0,$$
>
> then we say that S is a **linearly dependent set** and that the vectors in S are **linearly dependent**.

Note: Let S be a finite set of vectors in $\mathbb{R}^p$ and A the matrix with the vectors in S as **columns** in order.

Then S is linearly independent if and only if $Ax = 0$ has the unique solution $x = 0$.

Basis and Dimension

> **Basis Definition** A set B of vectors in a vector space V is called a **basis** for V if B is linearly independent and $V = \operatorname{span}(B)$.

Theorem (Dimension Exists): If a vector space V has a finite spanning set then it has a finite basis and every basis of V has the same number of elements.

In this case, we call the number of vectors in a basis of V the **dimension** of V and write it as $\dim(V)$.

Standard basis: For $\mathbb{R}^n$ or $\mathbb{C}^n$ the **standard basis**, usually denoted $\mathcal{S}$ is the set of vectors $\{e_1, ..., e_n\}$ where e_i has all entries zero except the ith entry which is 1. If follows that $\dim \mathbb{R}^n = n$.

For $\mathbb{P}_n$, the **standard basis**, also usually denoted $\mathcal{S}$, is the set of polynomials $\{1, t, t^2, ..., t^n\}$. $\dim \mathbb{P}_n = n + 1$

Theorem: Let V be a vector space. If S is a finite spanning set of V and T is a linearly independent set in V, then

$$|T| \leq \dim(V) \leq |S|$$

Furthermore
- if $|T| = \dim(V)$, T is a basis for V
- if $|S| = \dim(V)$, S is a basis for V

Coordinate Vectors

> **Coordinate Vectors** Suppose that $B = \{b_1, b_2, ..., b_n\}$ is a basis for a vector space V where we fix the order of these vectors.
>
> Let v be a vector in V. The fact that B spans V tell us that
>
> $$v = x_1 b_1 = x_2 b_2 + ... + x_n b_n$$
>
> for certains scalars $x_1, x_2, ..., x_n$; the fact that B is linearly independent tells us that for any specific v, there is only **ony** possible choice of these scalars.
>
> The scalars x_i are called the **coordinates** of the vector v with respect to the basis B.

Linear Maps

> **Linear Map** Let V and W be two vector spaces over the same field $\mathbb{F}$. A function $T : V \to W$ is called a **linear map**, a **linear tranformation** or described as **linear** if the following two conditions are satisfied:
>
> - Addition is respected
>
> $T(u + v) = T(u) + T(v) \text{ for all } u, v \in V$
> - Scalar multiplication is repected
>
> $T(\lambda v) = \lambda T(v) \text{ for all } \lambda \in \mathbb{F} \land v \in V$

> **Linearity Test Theorem** A function $T : V \to W$ is a linear map if and only if for all $\lambda, \mu \in \mathbb{F}$ and $u, v \in V$
>
> $$T(\lambda u + \mu v) = \lambda T(u) + \mu T(v)$$
>
> If $T(0) \neq 0$, or there is a vector v with $T(-v) \neq -T(v)$ then T is not linear.

Matrices of Linear Tranformations

Linear tranformations are very close and have an important relationship to matrix multiplication. e.g.

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 7x_1 + x_2 \\ x_2 + 4x_3 \end{pmatrix}$$

$$= \begin{pmatrix} 7 & 1 & 0 \\ 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

We say that $A = \begin{pmatrix} 7 & 1 & 0 \\ 0 & 1 & 4 \end{pmatrix}$ is the matrix of T with respect to standard bases.

Essentially

$$T(x) = Ax.$$

Matrix of a Linear Map Let $T : V \to W$ be a linear tranformation, where the domain V and the codomain W are finite-dimensional vector space over field $\mathbb{F}$.

Then A is the matrix of T with respect to ordered bases B for V and C for W if

$$[T(v)]_C = A[v]_B$$

for all vectors $v \in V$.

Here $[v]_B$ denotes the coordinate vector of v with respect to the basis B, and $[T(v)]_C$ denotes the coordinate vector of $T(v)$ with respect to the basis C.

A transformation matrix $A \in M_{q,p}$ is a linear map from a p dimensional vector space to a q dimensional has a $q \times p$ matrix.

Alternative method of finding the matrix of a linear transformation:
- Calculate the $T(v)$ for all v in the basis vector.
- Combine each transformed vector columnwise to find the transformation matrix.

Commutative Diagrams and Matricies of Linear Transformations

Lemma 1 If P is the matrix for $T_1 : U \to V$ with respect to bases B and C' and Q is the matrix for $T_2 : V \to W$ with respect to bases C' and C. Then QP is the matrix for $T_2 \circ T_1$ with repect to bases B and C.

Lemma 2 Let $T : V \to W$ be a linear transformation where $\dim(V) = \dim(W)$. Suppose M is the matrix of T with repect to bases B in V and C in W.

Then T is a bijection if and only if M is invertible, and in that case M^{-1} is the matrix of $T^{-1} : W \to V$ with repect to the bases C in W and B in V.

$$\xrightarrow[\text{Matrix}]{\text{Linear map}}$$

$$\begin{tikzcd}
V \text{ w.r.t } \mathcal{S}_V \arrow[r, "T", "A"'] & W \text{ w.r.t } \mathcal{S}_W \\
V \text{ w.r.t } B \arrow[r, "T", "M"'] \arrow[u, "\mathrm{id}_V", "P"'] & W \text{ w.r.t } C \arrow[u, "\mathrm{id}_W", "Q"']
\end{tikzcd}$$

We call this a **commutative** diagram because you get the same result whichever path we take from one corner to another.

e.g.

$$\begin{aligned} \mathrm{id}_W \circ T &= T \circ \mathrm{id}_V \\ T &= Q^{-1} \circ T \circ P \end{aligned}$$

by the two lemmas, this means that

$$QM = AP$$
$$M = Q^{-1}AP$$

w.r.t with respect to the basis

Kernal and Image

Kernal and Image of a Linear Transformation Let $T : V \to W$ be a linear map.

The **kernal** (or **nullspace**) of T, written $\ker(T)$, is the set of all zeros of T, that is,

$$\ker(T) = \{v \in V : T(v) = 0\}.$$

The **image** of T is the set of all function values of T, that is,

$$\text{im}(T) = \{w \in W : w = T(v) \text{ for some } v \in V\}.$$

Kernal and Image of a Matrix The **kernal** of a $p \times q$ matrix A is the subset of $\mathbb{R}^q$ defined by

$$\ker(A) - \{x \in \mathbb{R}^q : Ax = 0\}.$$

The **image** (or **nullspace**) of a $p \times q$ matrix A is the subset of $\mathbb{R}^p$ defined by

$$\text{im}(A) = \{b \in \mathbb{R}^p : b = Ax \text{ for some } x \in \mathbb{R}^q\}.$$

Rank and Nullity The **rank** of a linear map T is the dimension of $\text{im}(T)$ and the **rank** of a matrix A is the dimension of $\text{im}(A)$, denoted by $\text{rank}(T)$ and $\text{rank}(A)$.

The **nullity** of a linear map T is the dimension of $\ker(T)$ and the **nullity** of a matrix A is the dimension of $\ker(A)$, denoted by $\text{nullity}(T)$ and $\text{nullity}(A)$.

Rank-Nullity Theorem If A is an $p \times q$ matrix, then $\text{rank}(A) + \text{nullity}(A) = q$.

If $T : V \to W$ is a linear map between finite dimensional vector spaces

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

Some Linear Transformations on $\mathbb{R}^2$ and $\mathbb{R}^3$

Transformation	**Matrix**
$\mathbb{R}^2$	
Rotation by θ	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
Reflection across x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection across y-axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
Reflection across $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Scaling by k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$
Horizontal shear by k	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$
Vertical shear by k	$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$

Transformation	**Matrix**
Projection onto x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
Projection onto y-axis	$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$
$\mathbb{R}^3$	
Rotation about z-axis	$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$
Rotation about x-axis	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$
Rotation about y-axis	$\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$
Reflection across xy-plane	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$
Scaling by k	$\begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}$
Projection onto xy-plane	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Statistics

Revision

Definitions

Random experiment any experiment where the outcome cannot be predicted

Outcome the result of a random experiment

Event a set of possible outcomes

Sample space the set containing all possible outcomes

Probability of an event E $P(E)$ is the number measuring how likely event E is to occur from a random experiment.

Union $A \cup B$ the probability that either A or B occurs

Intersection $A \cap B$ the probability both A and B occur

Complement A^C the probability that A does not occur

$A \subseteq B$ A implies B

DeMorgan's $(A \cup B)^C = A^C \cap B^C$ and $\left(A^C \cap B^C\right) = (A \cup B)^C$

Probability Rules

- If $A \cap B = \emptyset$, A and B are mutually exclusive events.
- For any event E, $0 \leq \mathbb{P}(E) \leq 1$
- $\mathbb{P}(S) = 1$
- For any sequence of mutually exclusive events E_n, $\mathbb{P}\left(\bigcup_{i=0}^{n} E_i\right) = \sum_{i=0}^{n} \mathbb{P}(E_i)$
- $\mathbb{P}\left(E^C\right) = 1 - \mathbb{P}(E)$
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$

Conditional Probability

- The conditional probability of B conditional on A is $\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}$
- If $B \subseteq A$, then $\mathbb{P}(A \mid B) = 1$
- $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \times \mathbb{P}(B)$
- Law of total probability: $\mathbb{P}(A) = \mathbb{P}(A \mid B) \times \mathbb{P}(B) + \mathbb{P}\left(A \mid B^C\right) \times \mathbb{P}\left(B^C\right)$
- Bayes' rules: If $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$ then $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A) \times \frac{\mathbb{P}(A)}{\mathbb{P}(B)}$

Event Independence

- Two events are independent iff $\mathbb{P}(A \cap B) = \mathbb{P}(A) \times \mathbb{P}(B)$
- Independence implies that $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ and $\mathbb{P}(B \mid A) = \mathbb{P}(B)$

Introduction and descriptive statistics

Statistics is the science of the collection, processing, analysis, and interpretation of data

Population Obtain information about the entire collection of elements

Sample A subset of the population

Only sampling scheme that guarantees the sample to be a representative of the population is **random sampling**. Information drawn in a non-random sample cannot be generalised to larger populations.

Graphical Summary

Variable Types	**Graphical Plot**
One Categorical	Histogram
One Quantitative	Boxplot / Histogram
Two Categorical	Clustered bar chart `crosstab`
One Categorical One Quantitative	Comparative boxplot
Two Quantitative	Scatterplot

Histogram:

- Symmetric (even)
  - Skewness ~ 0
- Left skewed (trailing tail to the left)
  - Median > mean
  - Negative skewness
- Right skewed (trailing tail to the right)
  - Median < mean
  - Positive skewness
- Bimodal (two peaks)

Numerical Summary

Location a value most of the sample is centred around (mean, median, mode …)
Variability how spread the values are around the centre

Location Measurements

Sample mean $\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

Sample median $\tilde{x} = \begin{cases} \frac{1}{2}\left(x_{\frac{n}{2}} + x_{\frac{n+1}{2}}\right) & \text{even} \\ x_{\frac{n+1}{2}} & \text{odd} \end{cases}$ is the middlemost value of the sample

Quartiles The value such that some percent of observations are less or than equal to the value.
(0% = min, 50% = median, 100% = max)

Five number summary 0%, 25%, 50%, 75%, 100% quartiles.

Variability Measurements
Sample variance $s^2 = \frac{1}{n-1} \sum_{i=0}^{n} (x_i - \overline{x})^2$
Sample Standard Deviation $s = \sqrt{s^2}$
IQR interquartile range = $q_3 - q_1$
an outlier is an observation farther than $1.5 \times$ iqr from the closest quartile.

Random Variables

> **Random Variable** A random variable is a real valued function defined over the sample space:
>
> $$X : S \to \mathbb{R}$$
> $$\omega \to X(\omega)$$

Usually, random variables are denoted by an uppercase letter.

> **Domain of Variation** The domain of variation of X, S_X, is the set of all possible values that X can take.

$\boldsymbol{X}$ the name of the random variable
$\boldsymbol{X(\omega)}$ the numerical value taken by the random variable at some sample point ω
$\boldsymbol{x}$ a generic numerical value

> **Cumulative Distribution Function** The cdf (cumulative distribution function) of a random variable X is defined for any real number x by
>
> $$F(x) = \mathbb{P}(X \leq x)$$

All probability questions about X can be answered in terms of its distribution.

Notation: $X \sim F$ reads X follows the distribution F.

- For any $a \leq b$, $\mathbb{P}(a < X \leq b) = F(b) - F(a)$

Discrete Random Variables

> **Discrete Random Variable** A random variable is **discrete** if it can only assume a finite (or at most countably infinite) number of values.

> **Probability Mass Function** The pmf of a discrete random variable X is defined for any real number x by
>
> $$p(x) = \mathbb{P}(X = x)$$

The **Bernoulli random variable** is the simplest possible discrete random variable. $S_X = \{0, 1\}$ where $\mathbb{P}(0) = \pi - 1$ and $\mathbb{P}(1) = \pi$.

Continuous Random Variable

> **Continuous Random Variable** A random variable X is **continuous** if there exists a nonnegative function $f(x)$ defined for all real $x \in \mathbb{R}$ such that for any set B of real numbers,

$$\mathbb{P}(X \in B) = \int_B f(x)\,\mathrm{d}x$$

$f(x)$ is the **probability density function** (pdf).

Pdf Properties

- $F(x) = \mathbb{P}(X \leq x) = \int_{-\inf}^{x} f(y)\,\mathrm{d}y$
- $f(x) \geq 0 \qquad \forall x \in \mathbb{R}$
- $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$
- $\int_{-\inf}^{+\inf} f(x)\,\mathrm{d}x = 1$

Expectation and variance

Expectation The **expectation** or the **mean** of a random variable X, denoted $\mathbb{E}(X)$ or μ is:

Discrete	Continuous
$\mu = \mathbb{E}(X) = \sum_{x \in S_X} x p(x)$	$\mu = \mathbb{E}(X) = \int_{S_X} x f(x)\,\mathrm{d}x$

The **expectation** has the same unit as X.

$\mathbb{E}(X)$ is a weighted average of all the possible values of X, with each value being weighted by the probability that X assumes it.

$\mathbb{E}(X)$ has the same units as X.

Expectation of a function of a random variable

Discrete	Continuous
$\mathbb{E}(g(X)) = \sum_{x \in S_X} g(x)p(x)$	$\mathbb{E}(g(X)) = \int_{S_X} g(x)f(x)\,\mathrm{d}x$

Variance The **variance** of a random variable X, usually denoted by $\mathbb{V}\mathrm{ar}(X) \vee \sigma^2$ is defined by

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}((X-\mu)^2)$$

Discrete	Continuous
$\sigma^2 = \mathbb{V}\mathrm{ar}(X) = \sum_{x \in S_X} (x-\mu)^2 p(x)$	$\sigma^2 = \mathbb{V}\mathrm{ar}(X) = \int_{S_X} (x-\mu)^2 f(x)\,\mathrm{d}x$

variance has the unit of X^2

Alternative formula:

$$\sigma^2 = \mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \mathbb{E}(X^2) - \mu^2$$

Standard Deviation The **standard deviation** or X, $\sigma = \sqrt{\sigma^2} = \sqrt{\mathbb{V}\mathrm{ar}(X)}$

Standardisation

Suppose you have a random variable X with mean μ and variance σ^2.

The associated **standardised** random variable, often denoted Z is given by

$$Z = \frac{X - \mu}{\sigma}$$

Thus:

$$\mathbb{E}(Z) = \frac{1}{\sigma}\mathbb{E}(X) - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0$$

$$\mathbb{V}\mathrm{ar}(Z) = \frac{1}{\sigma^2}\mathbb{V}\mathrm{ar}(X) = \frac{\sigma^2}{\sigma^2} = 1$$

A standardised random varable always has **mean 0** and **variance 1**.

Jointly Distributed Random Variables

Joint Cumulative Distribution Function The **joint cumulative distribution function** of X and Y is given by

$$F_{XY}(x, y) = \mathbb{P}(X \leq x, Y \leq y) \quad \forall (x, y) \in \mathbb{R} \times \mathbb{R}$$

Joint Probability Mass Function If X and Y are both discrete, the **joint probability mass function** is defined by

$$p_{XY}(x, y) = \mathbb{P}(X = x, Y = y)$$

The **marginal pmf** of X and Y can be obtained by

$$p_{X(x)} = \sum_{y \in S_Y} p_{XY}(x, y)$$
$$p_{Y(y)} = \sum_{x \in S_X} p_{XY}(x, y)$$

Jointly Continuous X and Y are said to be **jointly continuous** if there exists a function $f_{XY}(x, y) : \mathbb{R} \times \mathbb{R} \to \mathbb{R}^+$ such that for any sets A and B of real number

$$\mathbb{P}(X \in A, Y \in B) = \int_A \int_B f_{XY}(x, y) \,\mathrm{d}y \,\mathrm{d}x$$

$f_{XY}(x, y)$ is the **joint probability density function** of X and Y.

The **marginal** densities follow from

$$\int_A f_{X(x)} \,\mathrm{d}x = \mathbb{P}(X \in A) = \mathbb{P}(X \in A, Y \in S_Y) = \int_A \int_{S_Y} f_{XY}(x, y) \,\mathrm{d}y \,\mathrm{d}x$$

Thus,

$$f_{X(x)} = \int_{S_Y} f_{XY}(x, y) \, \mathrm{d}y$$

$$f_{Y(y)} = \int_{S_X} f_{XY}(x, y) \, \mathrm{d}x$$

The **expectation of a function of two random variables** $g : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is

$$\begin{aligned} \mathbb{E}(g(X, Y)) &= \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) p_{XY}(x, y) \\ &= \int_{S_X} \int_{S_Y} g(x, y) f_{XY}(x, y) \, \mathrm{d}y \, \mathrm{d}x \end{aligned}$$

Independent Random Variables

Independent Random Variables The random variables X and Y are said to be **independent** is, for all $(x, y) \in \mathbb{R} \times \mathbb{R}$,

$$\mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(X \leq x) \times P(Y \leq y)$$

If X and Y are **independent**, then for any functions h and g

$$\mathbb{E}(h(X)g(Y)) = \mathbb{E}(h(X)) \times \mathbb{E}(g(Y))$$

Covariance and Correlation

Covariance of two random variables The **covariance** of two random variables X and Y is defined by

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}((X - \mathbb{E}(X))(Y - \mathbb{E}(Y)))$$

Properties
- $\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{C}\mathrm{ov}(Y, X)$
- $\mathbb{C}\mathrm{ov}(X, X) = \mathbb{V}\mathrm{ar}(X)$
- $\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$
- $\mathbb{C}\mathrm{ov}(aX + b, cY + d) = ac\mathbb{C}\mathrm{ov}(X, Y)$
- $\mathbb{C}\mathrm{ov}(X_1 + X_2, Y_1 + Y_2) = \mathbb{C}\mathrm{ov}(X_1, Y_1) + \mathbb{C}\mathrm{ov}(X_1, Y_2) + \mathbb{C}\mathrm{ov}(X_2, Y_1) + \mathbb{C}\mathrm{ov}(X_2, Y_2)$

Intepretation
- If $\mathbb{C}\mathrm{ov}(X, Y) > 0$, X and Y tend to increase or decrease together.
- If X and Y are independent $\Rightarrow \mathbb{C}\mathrm{ov}(X, Y) = 0$, however $\mathbb{C}\mathrm{ov}(X, Y) = 0 \not\Rightarrow X$ and Y independent.
- $\mathbb{C}\mathrm{ov}(X, Y) = 0 \Rightarrow$ no linear association between X and Y.
- If $\mathbb{C}\mathrm{ov}(X, Y) < 0$, X tends to increase as Y decrease and vice-versa.

Variance of sum of r.v

$$\mathbb{V}\mathrm{ar}(aX + bY) = a^2\mathbb{V}\mathrm{ar}(X) + b^2\mathbb{V}\mathrm{ar}(Y) + 2ab\mathbb{C}\mathrm{ov}(X, Y)$$

Correlation

Correlation Coefficient Covariance is important as an indicator of relationship between r.v's.

However, it depends on units of X and Y.

The **correlation coefficient** is often used instead.

$$\rho = \frac{\mathbb{C}\mathrm{ov}(X, Y)}{\sqrt{\mathbb{V}\mathrm{ar}(X)\mathbb{V}\mathrm{ar}(Y)}}$$

- It is dimensionless.
- Always between -1 and 1.
- Positive (and negative) ρ means positive (and negative) linear relationship.
- Closer $|\rho|$ is to 1, stronger the linear relationship.

Special Random Variables

Binomial Distribution